

AVINASH UMMAGANI

Software Engineer | Android
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SUMMARY

Software Engineer with 5 years of Android and AOSP experience. Specializes in Kotlin, Java, application architecture, real-time communication, and device integration, with end-to-end feature ownership and experience improving memory efficiency and system reliability.

PROFESSIONAL EXPERIENCE

Associate Software Engineer, Caavo Labs India Pvt Ltd

Apr 2022 - Present

Bengaluru, India (Remote) | Remote Caregiving System

- **Architecture and reliability:** Split a monolithic application into separate AOSP system APKs with Soong/Gradle builds, isolating failures to prevent cascading outages.
- **IPC and shared state:** Enabled cross-app data sharing through a content provider-based state service; implemented AIDL, broadcast receivers, and background/foreground services.
- **System apps and inputs:** Owned Android 11 AOSP onboarding apps through release; implemented Leanback focus/navigation, integrated three TV Input Framework inputs, and enabled IR, IP, CEC, and Bluetooth device control.
- **Real-time communication:** Integrated Zoom and OpenTok calling; built WebRTC from source with custom audio handling for remote control, and synchronized states across devices using Socket.IO.
- **Memory and stability:** Reduced memory use on 3 GB RAM devices through lazy allocation, buffer reuse, and disk-backed storage; verified a WebRTC leak fix that resolved allocation failures requiring device restarts.
- **Connectivity and recovery:** Owned Wi-Fi selection, connection, and recovery workflows; implemented OkHttp retry interceptors and connectivity recovery following system-wide network outages.
- **Device integration:** Implemented smart-plug setup and TV power-state detection using power readings reported over BLE/MQTT, with HDMI-CEC as a fallback.
- **State and media:** Implemented ViewModel/LiveData state management and Room/SQLite persistence; delivered cloud-backed photos and video streaming with Glide and ExoPlayer.
- **Notifications and diagnostics:** Delivered TV-friendly notifications, network-aware batched analytics uploads, and on-device bug reporting; integrated Firebase Crashlytics for production diagnostics.
- **Platform integration and review:** Integrated privileged apps into AOSP system images, configured permission allowlists, and handled runtime permissions; reviewed code through Gerrit with platform and hardware teams.

Systems Engineer, Tata Consultancy Services (TCS)

Jul 2021 - Apr 2022

Hyderabad, India | Enterprise CRM application

- **Feature delivery and collaboration:** Delivered features and resolved defects in an enterprise CRM application, collaborating with QA and backend teams through Agile sprints.

TECHNICAL SKILLS

Languages: Kotlin, Java, Python

Android and TV: Android SDK, Android TV, AOSP, Jetpack Compose, Leanback, WorkManager

Architecture and Dependency Injection: MVVM, Multi-module Architecture, Hilt, Dagger 2

Networking and Concurrency: Retrofit, OkHttp, Socket.IO, MQTT, Kotlin Coroutines, Flow, Channels

Data and Persistence: Room, SQLite, DataStore

Media and Communication: WebRTC, ExoPlayer

Device Connectivity: Bluetooth Low Energy (BLE), Wi-Fi, HDMI-CEC

Testing and Diagnostics: Appium, Robot Framework, Android Profiler, ADB, Logcat, Crashlytics, LeakCanary

Build and Version Control: Gradle, Soong, Repo, Jenkins, R8/ProGuard, Git, Gerrit

EDUCATION

B.Tech, Electronics & Communication Engineering

2021

VNR Vignana Jyothi Institute of Engineering & Technology, Hyderabad | CGPA: 9.5/10